

$$\begin{aligned}
& c_2 c_\gamma^{-2} (g_1^2 + g_2^2) + \\
& \frac{1}{16 \pi^2} \left(\frac{1}{48} \frac{1}{m_2} \left(3 g_1^4 \left(2 m_2^2 (-2 + c_2 \gamma^2) + C_H^2 S_4 \gamma^2 \right) 6 g_1^2 g_2^2 \left(4 m_2^2 (-1 + c_2 \gamma^2) + C_H^2 S_4 \gamma^2 \right) + \right. \right. \\
& \left. g_2^4 \left(3 C_H^2 S_4 \gamma^2 + 2 m_2^2 (-18 + c_2 \gamma^2 (4 + 9 c_2^2 + 9 S_2^2)) \right) \right) + \\
& \frac{4}{9} C_H^2 g_2^4 L F_{3,0}[m_2] + \frac{2}{3} C_H^2 g_2^4 L F_{4,-1}[m_2] - \frac{32}{45} C_H^2 g_2^4 L F_{5,-2}[m_2] + \\
& \frac{1}{4} \sum p S_2 \gamma S_4 \gamma g_1^2 \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) L F_{1,0}[m_d^p] - \frac{1}{12} \sum p g_1^4 c_2 \gamma^2 L F_{2,0}[m_d^p] + \\
& \frac{3}{4} S_2 \gamma S_4 \gamma \frac{1}{m_2} \overline{y_d^p} y_d^p (g_1^2 + g_2^2) (-C_H^2 + m_2^2) L F_{1,0}[m_d^r] + c_2 \gamma g_1^2 c_2 \gamma^2 \overline{y_d^p} y_d^p L F_{2,0}[m_d^r] + \\
& \frac{1}{4} \sum p S_2 \gamma S_4 \gamma g_1^2 \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) L F_{1,0}[m_e^p] - \frac{1}{4} \sum p g_1^4 c_2 \gamma^2 L F_{2,0}[m_e^p] + \\
& \frac{1}{4} S_2 \gamma S_4 \gamma \frac{1}{m_2} \overline{y_e^p} y_e^p (g_1^2 + g_2^2) (-C_H^2 + m_2^2) L F_{1,0}[m_e^r] + c_2 \gamma g_1^2 c_2 \gamma^2 \overline{y_e^p} y_e^p L F_{2,0}[m_e^r] + \\
& \frac{1}{4} S_2 \gamma S_4 \gamma \frac{1}{m_4^2} (g_1^2 + g_2^2) (-C_H^2 + m_2^2) (\overline{y_e^p} y_e^p + \sum p g_1^2) L F_{1,0}[m_l^p] - \\
& \frac{1}{9} c_2 \gamma (4 c_2^2 \overline{y_e^p} y_e^p (g_1^2 - g_2^2) + \sum p c_2 \gamma (g_1^4 + g_2^4)) L F_{2,0}[m_l^p] + \\
& \frac{1}{18} C_H^2 g_2^2 (36 c_2 \gamma c_\gamma^2 \overline{y_e^p} y_e^p + \sum p g_2^2 (4 - 9 c_2^2)) L F_{3,0}[m_l^p] + \frac{1}{12} C_H^2 g_2^2 \\
& (-24 c_2 \gamma c_\gamma^2 \overline{y_e^p} y_e^p + \sum p g_2^2 (-5 + 6 c_2^2)) L F_{4,-1}[m_l^p] + \frac{45}{8} \sum p C_H^2 g_2^4 L F_{5,-2}[m_l^p] + \\
& \frac{1}{4} S_2 \gamma S_4 \gamma \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) (-3 \overline{y_d^p} y_d^p + 3 \overline{y_u^p} y_u^p + \sum p g_1^2) L F_{1,0}[m_q^p] + \\
& \frac{1}{24} c_2 \gamma (12 c_2 \gamma^2 \overline{y_d^p} y_d^p (g_1^2 + 3 g_2^2) + 12 S_2^2 \overline{y_u^p} y_u^p (g_1^2 - 3 g_2^2) - \sum p c_2 \gamma (g_1^4 + 9 g_2^4)) \\
& L F_{2,0}[m_q^p] + \frac{1}{6} C_H^2 g_2^2 (36 c_2 \gamma (c_\gamma^2 \overline{y_d^p} y_d^p - S_2^2 \overline{y_u^p} y_u^p) + \sum p g_2^2 (4 - 9 c_2^2)) L F_{3,0}[m_q^p] + \\
& \frac{1}{4} C_H^2 g_2^2 (24 C_H^2 L F_{4,-2}[m_q^p] - c_2 \gamma \overline{y_d^p} y_d^p + S_2^2 \overline{y_u^p} y_u^p) + \sum p g_2^2 (-5 + 6 c_2^2)) L F_{4,-1}[m_q^p] + \\
& \frac{8}{15} \sum p C_H^2 g_2^4 L F_{5,-2}[m_q^p] + \frac{1}{2} \sum p S_2 \gamma S_4 \gamma g_1^2 \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) L F_{1,0}[m_u^p] - \\
& \frac{1}{3} \sum p g_1^4 c_2 \gamma^2 L F_{2,0}[m_u^p] + \frac{3}{4} S_2 \gamma S_4 \gamma \frac{1}{m_2} \overline{y_u^p} y_u^p (g_1^2 + g_2^2) (C_H^2 - m_2^2) L F_{1,0}[m_u^r] - \\
& 2 c_2 \gamma g_1^2 S_2^2 \overline{y_u^p} y_u^p L F_{2,0}[m_u^r] + \frac{1}{16} \frac{1}{m_4^2} S_4 \gamma^2 (C_H^2 - m_2^2) L F_{1,0}[m_a] (g_1^2 + g_2^2)^2 - \\
& \frac{3}{16} S_4 \gamma^2 L F_{1,1}[m_a] (g_1^2 + g_2^2)^2 + \frac{3}{16} C_H^2 S_4 \gamma^2 L F_{1,2}[m_a] (g_1^2 + g_2^2)^2 + \\
& \frac{1}{64} (2 g_1^2 g_2^2 (3 + c_4 \gamma (2 + 5 c_4 \gamma) - 4 S_2 \gamma^4) - g_2^4 (9 + c_4 \gamma (-6 + 5 c_4 \gamma) + 4 S_2 \gamma^4) - \\
& g_2^4 (1 + c_4 \gamma (2 + 5 c_4 \gamma) + 4 S_2 \gamma^4)) L F_{2,0}[m_a] + \frac{1}{288} C_H^2 (-9 g_1^4 ((-1 + c_4 \gamma)^2 + 4 S_2 \gamma^4) - \\
& 18 g_1^2 g_2^2 (-3 + c_4 \gamma (2 + c_4 \gamma) + 4 S_2 \gamma^4) - g_2^4 (17 + 9 c_4 \gamma (6 + c_4 \gamma) + 36 S_2 \gamma^4)) L F_{3,0}[m_a] + \\
& \frac{1}{96} C_H^2 (3 g_1^4 ((-1 + c_4 \gamma)^2 + 4 S_2 \gamma^4) + 6 g_1^2 g_2^2 (-3 + c_4 \gamma (2 + c_4 \gamma) + 4 S_2 \gamma^4) + \\
& g_2^4 (-13 + 3 c_4 \gamma (6 + c_4 \gamma) + 12 S_2 \gamma^4)) L F_{4,-1}[m_a] + \\
& \frac{45}{8} C_H^2 g_2^4 L F_{5,-2}[m_a] + \frac{2}{9} C_H^2 g_2^4 L F_{3,0}[\tilde{\mu}] + \frac{1}{3} C_H^2 g_2^4 L F_{4,-1}[\tilde{\mu}] - \frac{45}{48} C_H^2 g_2^4 L F_{5,-2}[\tilde{\mu}] + \\
& \frac{1}{2} m_1 S_4 \gamma \tilde{\mu} g_1^2 \frac{1}{m_4^2} (g_1^2 + g_2^2) (-C_H^2 + m_2^2) (c_\gamma^2 - S_2^2) L F_{1,1,0}[m_1, \tilde{\mu}] + \frac{1}{2} g_1^4 L F_{2,2,-2}[m_1, \tilde{\mu}] + \\
& \frac{1}{2} g_1^4 (m_1^2 + 8 m_1 S_2 \gamma \tilde{\mu} c_\gamma + 4 S_2^2 \tilde{\mu}^2 c_\gamma^2) L F_{2,2,-1}[m_1, \tilde{\mu}] + 2 g_1^4 m_1^2 S_2^2 \tilde{\mu}^2 c_\gamma^2 L F_{2,2,0}[m_1, \tilde{\mu}] + \\
& \frac{3}{2} m_2 S_4 \gamma \tilde{\mu} g_2^2 \frac{1}{m_4^2} (g_1^2 + g_2^2) (-C_H^2 + m_2^2) (c_\gamma^2 - S_2^2) L F_{1,1,0}[m_2, \tilde{\mu}] - \frac{8}{3} C_H^2 g_2^2 L F_{2,1,0}[m_2, \tilde{\mu}] + \\
& \frac{1}{2} g_2^4 (5 c_\gamma^4 + 2 S_2^2 c_\gamma^2 + 5 S_4^4) L F_{2,2,-2}[m_2, \tilde{\mu}] + \frac{1}{6} g_2^4 (3 m_2^2 (c_\gamma^4 + 10 S_2^2 c_\gamma^2 + S_4^4) - \\
& 2 C_H^2 (9 c_\gamma^4 + 8 S_2^2 + 9 S_4^4 + c_\gamma^2 (8 - 24 S_2^2)) - 72 m_2 S_2 \gamma \tilde{\mu} c_\gamma + 36 S_2^2 \tilde{\mu}^2 c_\gamma^2) L F_{2,2,-1}[m_2, \tilde{\mu}] - \\
& \frac{2}{3} m_2 S_2 \gamma c_\gamma g_2^4 (-9 m_2 S_2 \gamma c_\gamma \tilde{\mu}^2 + C_H^2 (6 m_2 S_2 \gamma c_\gamma + 8 \tilde{\mu}^2)) L F_{2,2,0}[m_2, \tilde{\mu}] + \\
& \frac{4}{3} C_H^2 g_2^4 L F_{3,1,-1}[m_2, \tilde{\mu}] + \frac{4}{3} C_H^2 g_2^4 (6 c_\gamma^4 + S_2^2 + 6 S_4^4 + c_\gamma^2 (1 - 12 S_2^2)) L F_{3,2,-2}[m_2, \tilde{\mu}] + \\
& \frac{2}{3} m_2 C_H^2 g_2^4 (-9 m_2 (c_\gamma^2 - S_2^2)^2 + 4 S_2 \gamma \tilde{\mu} c_\gamma) L F_{3,2,-1}[m_2, \tilde{\mu}] - \\
& 4 C_H^2 g_2^4 L F_{3,3,-3}[m_2, \tilde{\mu}] (c_\gamma^2 - S_2^2)^2 + 4 C_H^2 g_2^4 m_2^2 L F_{3,3,-2}[m_2, \tilde{\mu}] (c_\gamma^2 - S_2^2)^2 - \\
& 4 C_H^2 g_2^4 L F_{4,2,-3}[m_2, \tilde{\mu}] (c_\gamma^2 - S_2^2)^2 + 4 C_H^2 g_2^4 m_2^2 L F_{4,2,-2}[m_2, \tilde{\mu}] (c_\gamma^2 - S_2^2)^2 - \\
& 3 c_\gamma^4 \overline{y_d^p} \overline{y_d^s} y_d^p y_d^s L F_{1,1,0}[m_d^r, m_d^t] + \frac{3}{4} S_4 \gamma \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) \\
& (\tilde{\mu} \overline{y_d^p} (a_d^p (-c_\gamma^2 + S_2^2) + 2 S_2 \gamma \tilde{\mu} c_\gamma y_d^p) + \overline{a_d^p} (-2 S_2 \gamma c_\gamma a_d^p + \tilde{\mu} y_d^p (-c_\gamma^2 + S_2^2))) \\
& L F_{1,1,0}[m_d^r, m_q^p] + c_2 \gamma g_1^2 (c_\gamma \overline{a_d^p} - S_2 \gamma \tilde{\mu} \overline{y_d^p}) (c_\gamma a_d^p - S_2 \gamma \tilde{\mu} y_d^p) L F_{2,1,0}[m_d^r, m_q^p] + \\
& C_H^2 c_2 \gamma g_1^2 (c_\gamma \overline{a_d^p} - S_2 \gamma \tilde{\mu} \overline{y_d^p}) (c_\gamma a_d^p - S_2 \gamma \tilde{\mu} y_d^p) L F_{2,2,0}[m_d^r, m_q^p] - \\
& c_\gamma^4 \overline{y_e^p} \overline{y_e^s} y_e^p y_e^s L F_{1,1,0}[m_e^r, m_e^t] + \frac{1}{4} S_4 \gamma \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) \\
& (\tilde{\mu} \overline{y_e^p} (a_e^p (-c_\gamma^2 + S_2^2) + 2 S_2 \gamma \tilde{\mu} c_\gamma y_e^p) + \overline{a_e^p} (-2 S_2 \gamma c_\gamma a_e^p + \tilde{\mu} y_e^p (-c_\gamma^2 + S_2^2))) \\
& L F_{1,1,0}[m_e^r, m_l^p] + c_2 \gamma g_1^2 (c_\gamma \overline{a_e^p} - S_2 \gamma \tilde{\mu} \overline{y_e^p}) (c_\gamma a_e^p - S_2 \gamma \tilde{\mu} y_e^p) L F_{2,1,0}[m_e^r, m_l^p] + \\
& C_H^2 c_2 \gamma g_1^2 (c_\gamma \overline{a_e^p} - S_2 \gamma \tilde{\mu} \overline{y_e^p}) (c_\gamma a_e^p - S_2 \gamma \tilde{\mu} y_e^p) L F_{2,2,0}[m_e^r, m_l^p] + \frac{1}{4} \frac{1}{m_4^2} (\tilde{\mu} \overline{y_e^p} \\
& (a_e^p (-2 S_2 \gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (-1 + c_2 \gamma) + g_2^2 (1 + c_2 \gamma)) + S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_2^2)) + \\
& 2 S_2 \gamma y_e^p (-S_4 \gamma C_H^2 c_\gamma (g_1^2 + g_2^2) + S_2 \gamma c_2 \gamma m_2^2 (g_1^2 (-1 + c_2 \gamma) + g_2^2 (1 + c_2 \gamma)))) + \\
& \overline{a_e^p} (2 c_\gamma a_e^p (c_2 \gamma c_\gamma m_2^2 (g_1^2 (1 + c_2 \gamma) + 3 g_2^2 (1 + c_2 \gamma)) + 3 S_4 \gamma S_\gamma C_H^2 (g_1^2 + g_2^2)) + \tilde{\mu} y_e^p \\
& (-2 S_2 \gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (-1 + c_2 \gamma) + g_2^2 (1 + c_2 \gamma)) + S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_2^2)))) \\
& L F_{2,1,0}[m_l^p, m_e^r] + \frac{1}{4} \frac{1}{m_4^2} (g_1^2 + g_2^2) (-2 c_\gamma \overline{a_e^p} a_e^p (c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma S_\gamma C_H^2) - \\
& \tilde{\mu} (-2 S_2 \gamma c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (c_\gamma^2 - S_2^2)) (\overline{y_e^p} a_e^p + y_e^p \overline{a_e^p}) + \\
& 2 S_2 \gamma \tilde{\mu}^2 \overline{y_e^p} y_e^p (S_4 \gamma C_H^2 c_\gamma - S_2 \gamma m_2^2 c_2 \gamma^2)) L F_{3,1,-1}[m_l^p, m_e^r] + C_H^2 c_2 \gamma \\
& (g_1^2 (-1 + c_2 \gamma) + g_2^2 (1 + c_2 \gamma)) (c_\gamma \overline{a_e^p} - S_2 \gamma \tilde{\mu} \overline{y_e^p}) (c_\gamma a_e^p - S_2 \gamma \tilde{\mu} y_e^p) L F_{3,1,0}[m_l^p, m_e^r] - \\
& C_H^2 c_2 \gamma g_1^2 (c_\gamma \overline{a_e^p} - S_2 \gamma \tilde{\mu} \overline{y_e^p}) (c_\gamma a_e^p - S_2 \gamma \tilde{\mu} y_e^p) L F_{3,2,-1}[m_l^p, m_e^r] - \frac{1}{2} C_H^2 (-1 + 2 c_2 \gamma) \\
& (2 g_2^2 + 3 c_2 \gamma (g_1^2 + g_2^2)) (c_\gamma \overline{a_e^p} - S_2 \gamma \tilde{\mu} \overline{y_e^p}) (c_\gamma a_e^p - S_2 \gamma \tilde{\mu} y_e^p) L F_{4,1,-1}[m_l^p, m_e^r] + \\
& \frac{2}{3} C_H^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) (c_\gamma \overline{a_e^p} - S_2 \gamma \tilde{\mu} \overline{y_e^p}) (c_\gamma a_e^p - S_2 \gamma \tilde{\mu} y_e^p) L F_{5,1,-2}[m_l^p, m_e^r] - \\
& c_\gamma^4 \overline{y_e^p} \overline{y_e^s} y_e^p y_e^s L F_{1,1,0}[m_l^p, m_l^s] - 2 C_H^2 c_\gamma^4 \overline{y_e^p} \overline{y_e^s} y_e^p y_e^s L F_{2,1,0}[m_l^p, m_l^s] + \\
& 2 C_H^2 c_\gamma^4 \overline{y_e^p} \overline{y_e^s} y_e^p y_e^s L F_{3,1,-1}[m_l^p, m_l^s] + \\
& \frac{1}{4} \frac{1}{m_4^2} (\tilde{\mu} \overline{y_d^p} (a_d^p (-2 S_2 \gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (1 + 3 c_2 \gamma) + 3 g_2^2 (1 + c_2 \gamma)) + \\
& 3 S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_2^2)) + \\
& 2 S_2 \gamma y_d^p (-3 S_4 \gamma C_H^2 c_\gamma (g_1^2 + g_2^2) + S_2 \gamma c_2 \gamma m_2^2 (g_1^2 (1 + 3 c_2 \gamma) + 3 g_2^2 (1 + c_2 \gamma)))) + \\
& \overline{a_d^p} (2 c_\gamma a_d^p (c_2 \gamma c_\gamma m_2^2 (g_1^2 (1 + 3 c_2 \gamma) + 3 g_2^2 (1 + c_2 \gamma)) + 3 S_4 \gamma S_\gamma C_H^2 (g_1^2 + g_2^2)) + \\
& \tilde{\mu} y_d^p (-2 S_2 \gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (1 + 3 c_2 \gamma) + 3 g_2^2 (1 + c_2 \gamma)) + \\
& 3 S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_2^2)))) L F_{2,1,0}[m_q^p, m_d^r] + \frac{3}{4} \frac{1}{m_4^2} (g_1^2 + g_2^2) \\
& (-2 c_\gamma \overline{a_d^p} a_d^p (c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma S_\gamma C_H^2) - \tilde{\mu} (-2 S_2 \gamma c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (c_\gamma^2 - S_2^2)) \\
& (\overline{y_d^p} a_d^p + y_d^p \overline{a_d^p}) - 2 S_2 \gamma \tilde{\mu}^2 \overline{y_d^p} y_d^p (S_4 \gamma C_H^2 c_\gamma - S_2 \gamma m_2^2 c_2 \gamma^2)) L F_{3,1,-1}[m_q^p, m_d^r] + \\
& C_H^2 c_2 \gamma (g_1^2 (1 + 3 c_2 \gamma) + 3 g_2^2 (1 + c_2 \gamma)) (c_\gamma \overline{a_d^p} - S_2 \gamma \tilde{\mu} \overline{y_d^p}) (c_\gamma a_d^p - S_2 \gamma \tilde{\mu} y_d^p) \\
& L F_{3,1,0}[m_q^p, m_d^r] - C_H^2 c_2 \gamma g_1^2 (c_\gamma \overline{a_d^p} - S_2 \gamma \tilde{\mu} \overline{y_d^p}) (c_\gamma a_d^p - S_2 \gamma \tilde{\mu} y_d^p) L F_{3,2,-1}[m_q^p, m_d^r] - \\
& \frac{3}{2} C_H^2 (-2 g_2^2 + c_2 \gamma (1 + 6 c_2 \gamma) (g_1^2 + g_2^2)) (c_\gamma \overline{a_d^p} - S_2 \gamma \tilde{\mu} \overline{y_d^p}) \\
& (c_\gamma a_d^p - S_2 \gamma \tilde{\mu} y_d^p) L F_{4,1,-1}[m_q^p, m_d^r] + \\
& 2 C_H^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) (c_\gamma \overline{a_d^p} - S_2 \gamma \tilde{\mu} \overline{y_d^p}) (c_\gamma a_d^p - S_2 \gamma \tilde{\mu} y_d^p) L F_{5,1,-2}[m_q^p, m_d^r] - \\
& 3 (c_\gamma^4 \overline{y_d^p} \overline{y_d^s} y_d^p y_d^s + S_4^4 \overline{y_u^p} \overline{y_u^s} y_u^p y_u^s) L F_{1,1,0}[m_q^p, m_q^s] - \\
& 6 C_H^2 (c_\gamma^2 \overline{y_d^p} y_d^s (c_\gamma^2 \overline{y_d^s} y_d^p - S_2^2 \overline{y_u^s} y_u^p) + S_4^4 \overline{y_u^p} \overline{y_u^s} y_u^p y_u^s) L F_{2,1,0}[m_q^p, m_q^s] + \\
& 6 C_H^2 (c_\gamma^2 \overline{y_d^p} y_d^s (c_\gamma^2 \overline{y_d^s} y_d^p - S_2^2 \overline{y_u^s} y_u^p) + S_4^4 \overline{y_u^p} \overline{y_u^s} y_u^p y_u^s) L F_{3,1,-1}[m_q^p, m_q^s] + \\
& \frac{3}{4} S_4 \gamma \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) (\tilde{\mu} \overline{y_u^p} (a_u^p (-c_\gamma^2 + S_2^2) - 2 S_2 \gamma \tilde{\mu} c_\gamma y_u^p) + \\
& \overline{a_u^p} (2 S_2 \gamma c_\gamma a_u^p + \tilde{\mu} y_u^p (-c_\gamma^2 + S_2^2))) L F_{1,1,0}[m_q^p, m_u^r] + \\
& \frac{1}{2} c_2 \gamma (g_1^2 - 3 g_2^2) (S_2 \gamma \overline{y_u^p} - \tilde{\mu} c_\gamma \overline{y_u^p}) (S_2 \gamma y_u^p - \tilde{\mu} c_\gamma y_u^p) L F_{2,1,0}[m_q^p, m_u^r] + \\
& \frac{1}{2} C_H^2 (c_2 \gamma g_1^2 + g_2^2 (4 - 3 c_2 \gamma)) (S_2 \gamma \overline{a_u^p} - \tilde{\mu} c_\gamma \overline{y_u^p}) (S_2 \gamma a_u^p - \tilde{\mu} c_\gamma y_u^p) L F_{2,2,0}[m_q^p, m_u^r] - \\
& 6 C_H^2 c_2 \gamma g_2^2 (S_2 \gamma \overline{a_u^p} - \tilde{\mu} c_\gamma \overline{y_u^p}) (S_2 \gamma a_u^p - \tilde{\mu} c_\gamma y_u^p) L F_{3,1,0}[m_q^p, m_u^r] + \\
& C_H^2 g_2^2 (-1 + 3 c_2 \gamma) (S_2 \gamma \overline{a_u^p} - \tilde{\mu} c_\gamma \overline{y_u^p}) (S_2 \gamma a_u^p - \tilde{\mu} c_\gamma y_u^p) L F_{3,2,-1}[m_q^p, m_u^r] + \\
& 6 C_H^2 c_2 \gamma g_2^2 (S_2 \gamma \overline{a_u^p} - \tilde{\mu} c_\gamma \overline{y_u^p}) (S_2 \gamma a_u^p - \tilde{\mu} c_\gamma y_u^p) L F_{4,1,-1}[m_q^p, m_u^r] + \\
& 6 C_H^2 S_2^2 c_\gamma^2 \overline{y_d^p} y_d^s \overline{y_u^s} y_u^p L F_{2,1,0}[m_q^s, m_q^p] - \\
& 6 C_H^2 S_2^2 c_\gamma^2 \overline{y_d^p} y_d^s \overline{y_u^s} y_u^p L F_{3,1,-1}[m_q^s, m_q^p] + \frac{1}{4} \frac{1}{m_4^2} (\tilde{\mu} \overline{y_u^p} \\
& (a_u^p (-2 S_2 \gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (-4 + 3 c_2 \gamma) + 3 c_2 \gamma g_2^2) + 3 S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_2^2)) + \\
& 2 \tilde{\mu} c_\gamma y_u^p (c_2 \gamma c_\gamma m_2^2 (g_1^2 (-4 + 3 c_2 \gamma) + 3 c_2 \gamma g_2^2) + 3 S_4 \gamma S_\gamma C_H^2 (g_1^2 + g_2^2))) + \\
& \overline{a_u^p} (2 S_2 \gamma a_u^p (-3 S_4 \gamma C_H^2 c_\gamma (g_1^2 + g_2^2) + S_2 \gamma c_2 \gamma m_2^2 (g_1^2 (-4 + 3 c_2 \gamma) + 3 c_2 \gamma g_2^2)) + \tilde{\mu} y_u^p \\
& (-2 S_2 \gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (-4 + 3 c_2 \gamma) + 3 c_2 \gamma g_2^2) + 3 S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_2^2)))) \\
& L F_{2,1,0}[m_u^r, m_q^p] + \frac{3}{4} \frac{1}{m_4^2} (g_1^2 + g_2^2) (2 S_2 \gamma \overline{a_u^p} a_u^p (S_4 \gamma C_H^2 c_\gamma - S_2 \gamma m_2^2 c_2 \gamma^2) + \\
& \tilde{\mu} (2 S_2 \gamma c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (-c_\gamma^2 + S_2^2)) (\overline{y_u^p} a_u^p + y_u^p \overline{a_u^p}) - \\
& 2 c_\gamma \tilde{\mu}^2 \overline{y_u^p} y_u^p (c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma S_\gamma C_H^2)) L F_{3,1,-1}[m_u^r, m_q^p] + \\
& C_H^2 (c_2 \gamma g_1^2 (-4 + 3 c_2 \gamma) + g_2^2 (-2 + 3 c_2 \gamma^2)) (S_2 \gamma \overline{a_u^p} - \tilde{\mu} c_\gamma \overline{y_u^p}) \\
& (S_2 \gamma a_u^p - \tilde{\mu} c_\gamma y_u^p) L F_{3,1,0}[m_u^r, m_q^p] - \\
& \frac{1}{2} C_H^2 (c_2 \gamma g_1^2 + g_2^2 (4 - 3 c_2 \gamma)) (S_2 \gamma \overline{a_u^p} - \tilde{\mu} c_\gamma \overline{y_u^p}) (S_2 \gamma a_u^p - \tilde{\mu} c_\gamma y_u^p) L F_{3,2,-1}[m_u^r, m_q^p] - \\
& 3 C_H^2 (c_2 \gamma g_1^2 (-2 + 3 c_2 \gamma) + g_2^2 (-2 + 3 c_2 \gamma^2)) \\
& (S_2 \gamma \overline{a_u^p} - \tilde{\mu} c_\gamma \overline{y_u^p}) (S_2 \gamma a_u^p - \tilde{\mu} c_\gamma y_u^p) L F_{4,1,-1}[m_u^r, m_q^p] + \\
& 2 C_H^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) (S_2 \gamma \overline{a_u^p} - \tilde{\mu} c_\gamma \overline{y_u^p}) (S_2 \gamma a_u^p - \tilde{\mu} c_\gamma y_u^p) L F_{5,1,-2}[m_u^r, m_q^p] - \\
& 3 S_2^2 \overline{y_u^p} \overline{y_u^s} y_u^p y_u^s L F_{1,1,0}[m_u^r, m_u^t] - \frac{4}{3} C_H^2 c_2 \gamma^2 (g_1^2 + g_2^2) L F_{2,1,-1}[\tilde{\mu}, m_1] + \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^2 \frac{1}{m_2^2} (g_1^2 + g_2^2) (-2 S_2 \gamma c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (c_\gamma^2 - S_2^2)) L F_{2,1,0}[\tilde{\mu}, m_1] + \\
& \frac{1}{2} g_1^2 c_2 \gamma^2 (g_1^2 + g_2^2) L F_{3,1,-2}[\tilde{\mu}, m_1] + \frac{1}{2} g_1^2 \frac{1}{m_2^2} (g_1^2 + g_2^2) \\
& (m_1 S_4 \gamma C_H^2 \tilde{\mu} (-c_\gamma^2 + S_2^2) + 2 m_2^2 c_2 \gamma^2 (-2 C_H^2 + m_1 S_2 \gamma \tilde{\mu} c_\gamma)) L F_{3,1,-1}[\tilde{\mu}, m_1] - \\
& 2 m_1 S_2 \gamma C_H^2 \tilde{\mu} c_\gamma g_1^2 c_2 \gamma^2 (g_1^2 + g_2^2) L F_{3,1,0}[\tilde{\mu}, m_1] + \frac{3}{2} C_H^2 g_1^4 L F_{3,2,-2}[\tilde{\mu}, m_1] + \\
& \frac{1}{2} C_H^2 g_1^4 (3 m_1^2 + 20 m_1 S_2 \gamma \tilde{\mu} c_\gamma + 8 S_2^2 \tilde{\mu}^2 c_\gamma^2) L F_{3,2,-1}[\tilde{\mu}, m_1] + \\
& 4 C_H^2 g_1^4 m_1^2 S_2^2 \tilde{\mu}^2 c_\gamma^2 L F_{3,2,0}[\tilde{\mu}, m_1] + \\
& \frac{1}{3} C_H^2 g_1^2 (-5 g_2^2 + 12 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{4,1,-2}[\tilde{\mu}, m_1] + \\
& 2 m_1 S_2 \gamma C_H^2 \tilde{\mu} c_\gamma g_1^2 (-g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{4,1,-1}[\tilde{\mu}, m_1] - \\
& 4 C_H^2 g_1^4 L F_{4,2,-3}[\tilde{\mu}, m_1] + C_H^2 g_1^4 (-m_1^2 - 8 m_1 S_2 \gamma \tilde{\mu} c_\gamma - 4 S_2^2 \tilde{\mu}^2 c_\gamma^2) L F_{4,2,-2}[\tilde{\mu}, m_1] - \\
& C_H^2 g_1^2 L F_{4,2,-1}[\tilde{\mu}, m_1] - \\
& \frac{2}{3} C_H^2 g_1^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{5,1,-3}[\tilde{\mu}, m_1] - \\
& \frac{4}{3} m_1 S_2 \gamma C_H^2 \tilde{\mu} c_\gamma g_1^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{5,1,-2}[\tilde{\mu}, m_1] - \\
& \frac{9}{4} g_2^2 c_2 \gamma^2 (g_1^2 + g_2^2) L F_{2,1,-1}[\tilde{\mu}, m_2] + \\
& \frac{3}{2} m_2 \tilde{\mu} g_2^2 \frac{1}{m_2^2} (g_1^2 + g_2^2) (-2 S_2 \gamma c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (c_\gamma^2 - S_2^2)) L F_{2,1,0}[\tilde{\mu}, m_2] + \\
& \frac{3}{2} g_2^2 c_2 \gamma^2 (g_1^2 + g_2^2) L F_{3,1,-2}[\tilde{\mu}, m_2] + \frac{1}{6} g_2^2 \frac{1}{m_2^2} (-4 C_H^2 m_2^2 (-8 g_2^2 + 9 c_2 \gamma^2 (g_1^2 + g_2^2)) + \\
& 9 m_2 \tilde{\mu} (g_1^2 + g_2^2) (2 S_2 \gamma c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (-c_\gamma^2 + S_2^2))) L F_{3,1,-1}[\tilde{\mu}, m_2] - \\
& \frac{2}{3} m_2 S_2 \gamma C_H^2 \tilde{\mu} c_\gamma g_2^2 (-8 g_2^2 + 9 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{3,1,0}[\tilde{\mu}, m_2] + \\
& \frac{1}{6} C_H^2 g_2^4 (69 c_\gamma^4 + 16 S_2^2 + 69 S_4^4 + c_\gamma^2 (16 - 6 S_2^2)) L F_{3,2,-2}[\tilde{\mu}, m_2] + \\
& \frac{1}{6} C_H^2 g_2^4 (3 m_2^2 (3 c_\gamma^4 + 38 S_2^2 c_\gamma^2 + 3 S_4^4) + 4 m_2 S_2 \gamma \tilde{\mu} c_\gamma (8 + 51 c_\gamma^2 + 51 S_2^2) + 72 S_2^2 \tilde{\mu}^2 c_\gamma^2) \\
& L F_{3,2,-1}[\tilde{\mu}, m_2] + 12 C_H^2 g_2^4 m_2^2 S_2^2 \tilde{\mu}^2 c_\gamma^2 L F_{3,2,0}[\tilde{\mu}, m_2] + \\
& 3 C_H^2 g_2^2 (-3 g_2^2 + 4 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{4,1,-2}[\tilde{\mu}, m_2] + \\
& 2 m_2 S_2 \gamma C_H^2 \tilde{\mu} c_\gamma g_2^2 (-7 g_2^2 + 9 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{4,1,-1}[\tilde{\mu}, m_2] - \\
& C_H^2 g_2^4 (7 c_\gamma^4 + 2 S_2^2 c_\gamma^2 + 7 S_4^4) L F_{4,2,-3}[\tilde{\mu}, m_2] - \\
& C_H^2 g_2^4 (m_2^2 (c_\gamma^4 + 14 S_2^2 c_\gamma^2 + S_4^4) + 32 m_2 S_2 \gamma \tilde{\mu} c_\gamma + 16 S_2^2 \tilde{\mu}^2 c_\gamma^2) L F_{4,2,-2}[\tilde{\mu}, m_2] - \\
& 16 C_H^2 g_2^4 m_2^2 S_2^2 \tilde{\mu}^2 c_\gamma^2 L F_{4,2,-1}[\tilde{\mu}, m_2] - 2 C_H^2 g_2^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{5,1,-3}[\tilde{\mu}, m_2] - \\
& 4 m_2 S_2 \gamma C_H^2 \tilde{\mu} c_\gamma g_2^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{5,1,-2}[\tilde{\mu}, m_2] + \\
& 8 C_H^2 g_1^2 g_2^2 L F_{2,2,-1,-2}[m_2, \tilde{\mu}] + \\
& 2 C_H^2 g_1^2 g_2^2 (3 m_1 m_2 + 2 S_2 \gamma \tilde{\mu} c_\gamma (7 m_1 + 5 m_2) + 12 S_2^2 \tilde{\mu}^2 c_\gamma^2) L F_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
& 16 m_1 m_2 C_H^2 g_1^2 g_2^2 S_2^2 \tilde{\mu}^2 c_\gamma^2 L F_{2,2,1,0}[m_2, \tilde{\mu}, m_1] - 4 C_H^2 g_1^2 g_2^2 L F_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] - \\
& 4 C_H^2 g_1^2 g_2^2 (m_1 m_2 + 4 S_2 \gamma \tilde{\mu} c_\gamma (m_1 + m_2) + 4 S_2^2 \tilde{\mu}^2 c_\gamma^2) L F_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] - \\
& 16 m_1 m_2 C_H^2 g_1^2 g_2^2 S_2^2 \tilde{\mu}^2 c_\gamma^2 L F_{3,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
& 6 c_\gamma^3 \overline{y_d^s} y_d^s (S_2 \gamma \tilde{\mu} \overline{y_d^p} a_d^p + \overline{a_d^p} (-c_\gamma a_d^$$